

Foundations of Osteopathic Medicine

Signal Transduction I: Ion channel-linked receptors and G-protein coupled receptors

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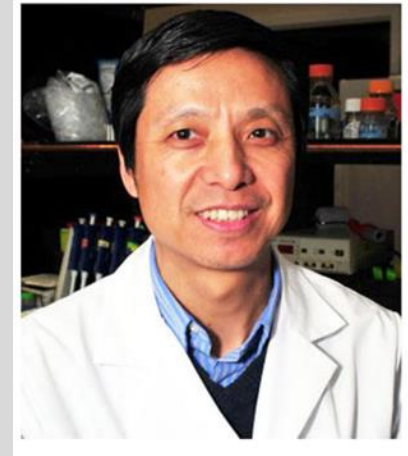
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Session Objectives

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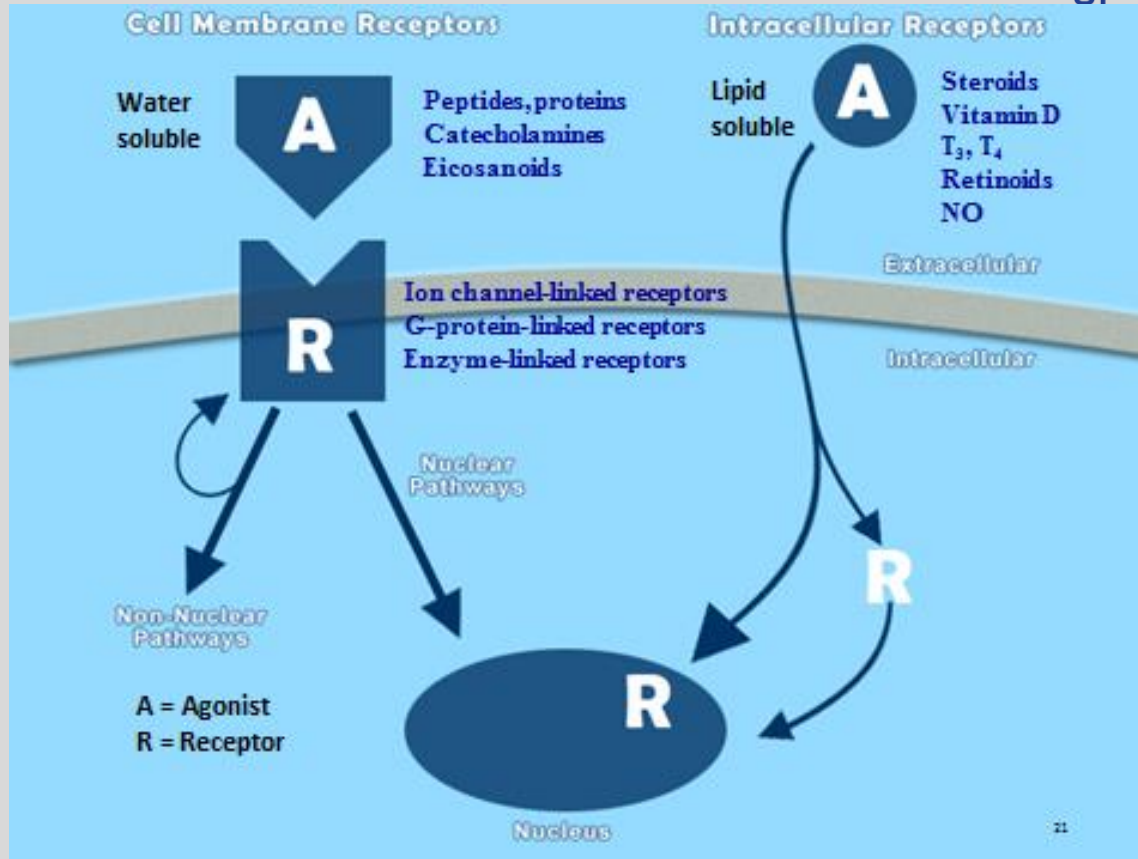
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1. Explain how ion channel-linked receptors work and give examples of such receptors
2. Discuss and be able to explain how heterotrimeric G protein-coupled receptors work
3. Describe the pathways which G proteins utilize involving the small-molecule second messengers Cyclic AMP, Calcium, Inositol trisphosphate (IP3) and Diacylglycerol (DAG)
4. Define the term eicosanoid and list the major classes of eicosanoid and the enzymes that make them

Receptors: Overview

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Transduction Pathways for Cell Membrane Receptors

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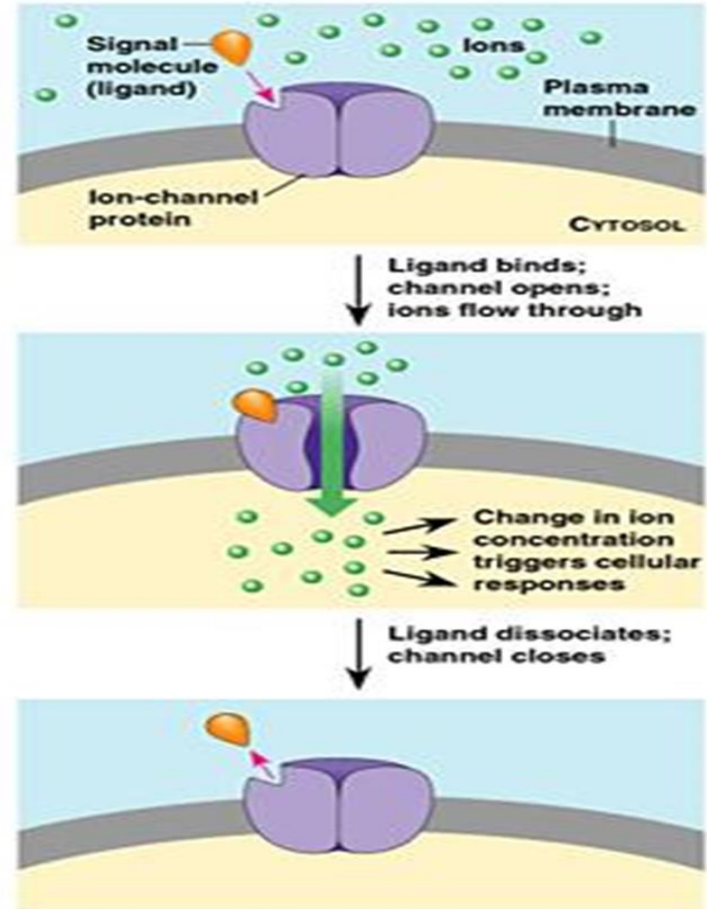
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- Ion channel-linked receptors, a.k.a “ligand-gated ion channels” or “ionotropic receptors”
- G-protein-linked receptors
- Enzyme-linked receptors, a.k.a. “catalytic receptors”

Ion Channel-linked Receptors

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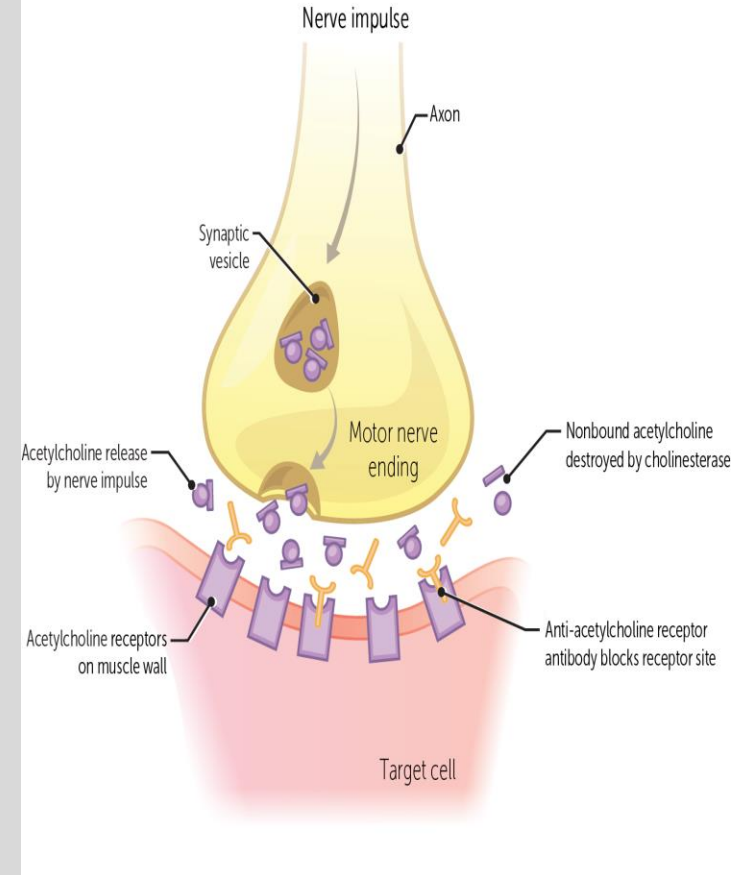
- Acts as gates in the cell membrane.
- Ligand binding opens the gate, which lets in specific ions through a channel in the receptor.
- The receptors are protein multimers that form the membrane-spanning pore in the center, known as central conducting pore.



Example:

Nicotinic acetylcholine (nACh) receptor

- The nACh receptor is an ion-channel-coupled receptor (In contrast, muscarinic ACh receptors are G protein coupled receptors, GPCRs).
- found at the neuromuscular junction and other parts of the nervous system
- ACh binding to the receptor increases Na^+ and K^+ permeability causing depolarization and the end-plate potential that opens voltage-gated Na^+ channels, which triggers action potential and muscular contraction
- Clinical Relevance:
 - The arrow poison Curare blocks binding of ACh to its receptor.
 - Myasthenia gravis (MG): autoantibodies to ACh receptor
 - Neostigmine & organophosphate insecticides inactivate acetylcholinesterase leading to the accumulation of ACh and overstimulation of its receptors.
 - Botulinum toxin, a bacterial poison that decreases the quantity of acetylcholine release by the nerve terminals

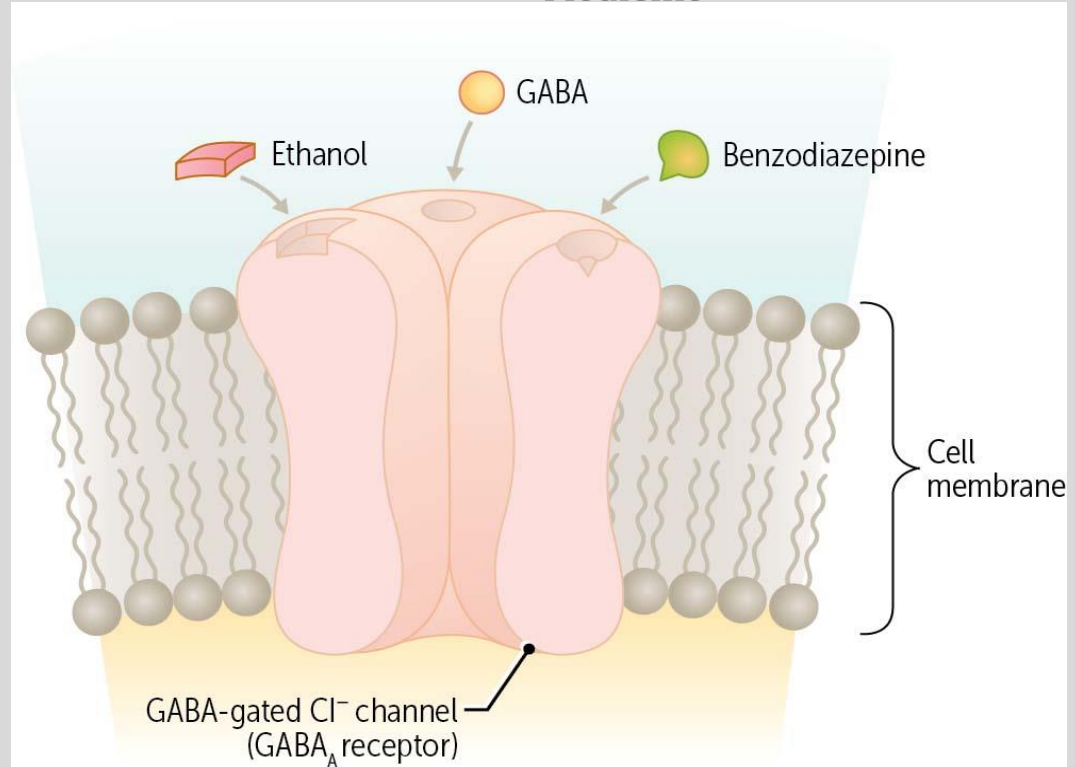


More examples of ion channel-linked receptors

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- IP3 receptor (ER; Ca^{+2} channel)
- Ryanodine receptor (ER; Ca^{+2} channel)
- ACh receptor superfamily: serotonin, GABA, glycine receptors



G protein-linked or coupled receptors (GPCR)

Why do you care about G protein-coupled receptors (GPCR) as a physician?

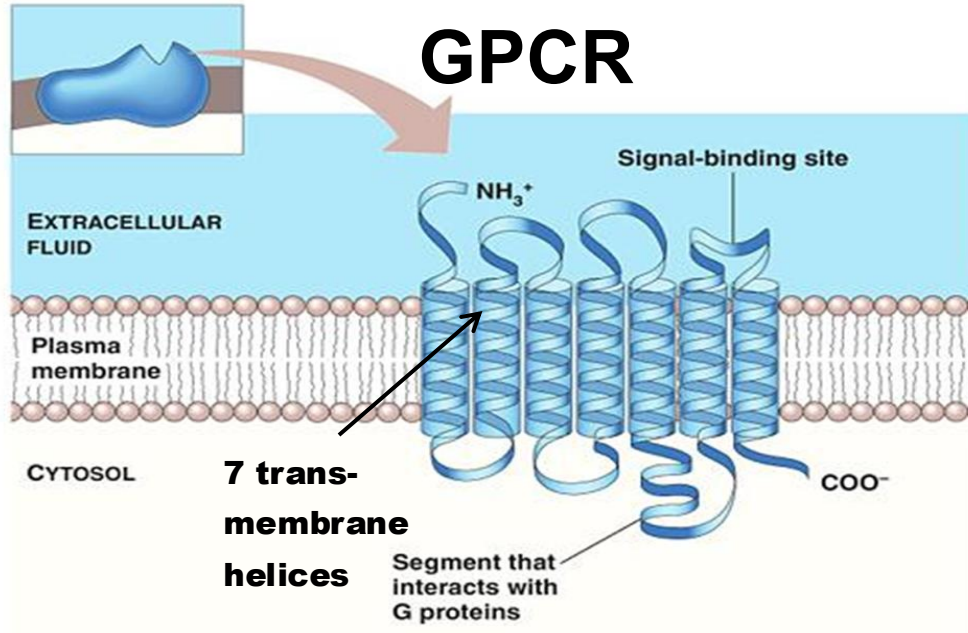
- The largest family of cell membrane receptors: >1000 GPCRs in humans
- Respond to numerous signals: hormones, neurotransmitters and light (photons).
- Mediate virtually every important physiological process, from immune function to taste and smell to the fight-or-flight response in humans.
- > One-third of all drugs approved by the FDA target GPCRs ([Nat Rev Drug Discov.](#) 2017 Dec;16:829-842).

G Protein-Coupled Receptors

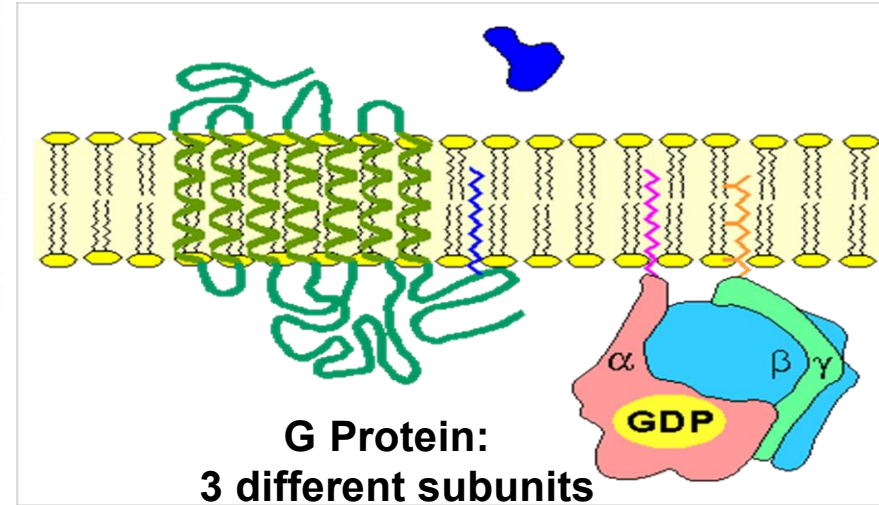
GPCR, cont.

- Naturally occurring small molecules which bind to GPCRs include adrenaline (epinephrine), dopamine, prostaglandins, and adenosine.
- Drug-like small molecules which bind to GPCRs include caffeine, morphine, heroin and histamine.
- Mutations in the GPCR genes are being increasingly recognized as important in the pathogenesis of various diseases
- GPCR are transmembrane proteins with seven membrane-spanning regions

Structure of GPCR and G proteins



GPCR + G Protein



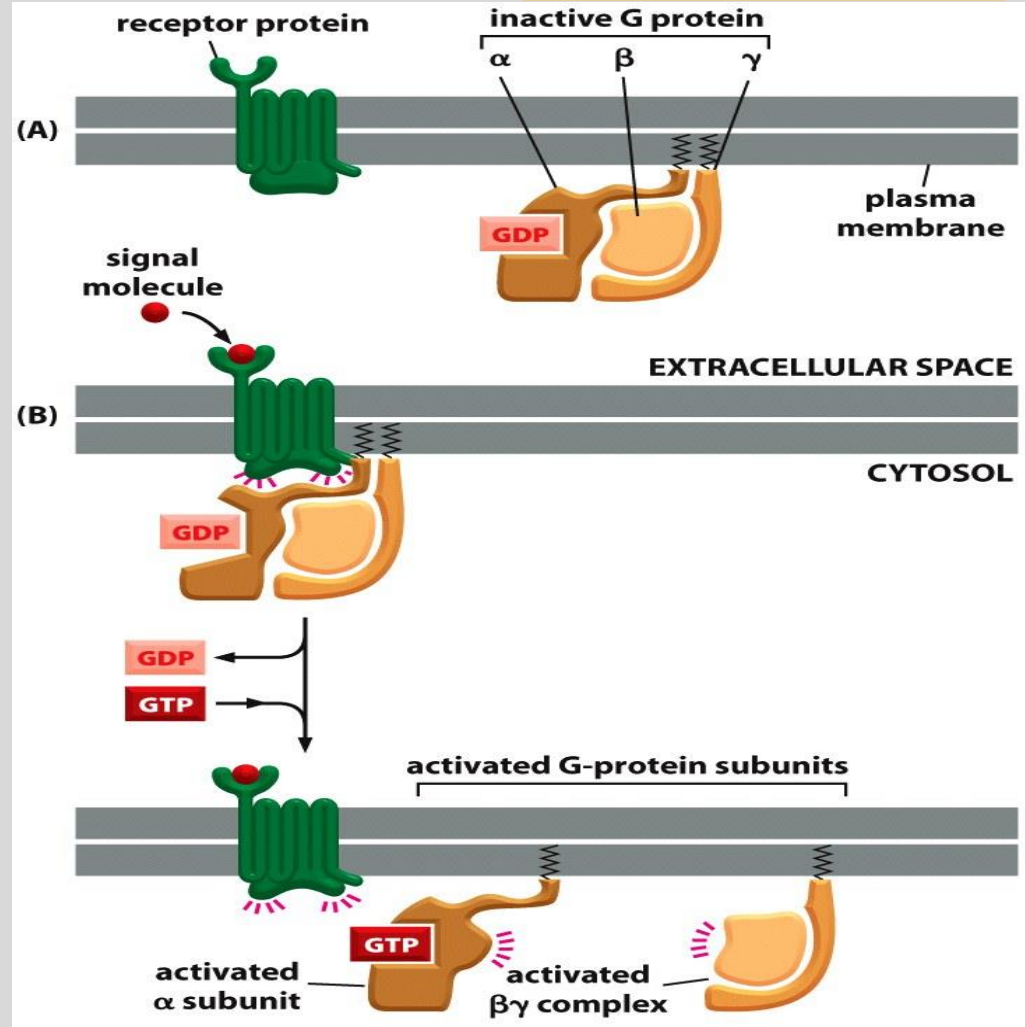
G Proteins: GTP/GDP-binding proteins

- Activated by the cell surface receptors and carry the signals from the receptors to intracellular effector proteins.
- Can (1) bind GTP, (2) hydrolyze GTP, and (3) switch between an active GTP-bound state and an inactive GDP-bound state.
- Have an intrinsic GTPase activity that hydrolyzes GTP to GDP, inactivating the G proteins.

G Protein: Activation & Inactivation

1. Agonist binding
2. Receptor-G-protein interaction
3. GDP release
4. GTP binding
5. α dissociates from $\beta\gamma$

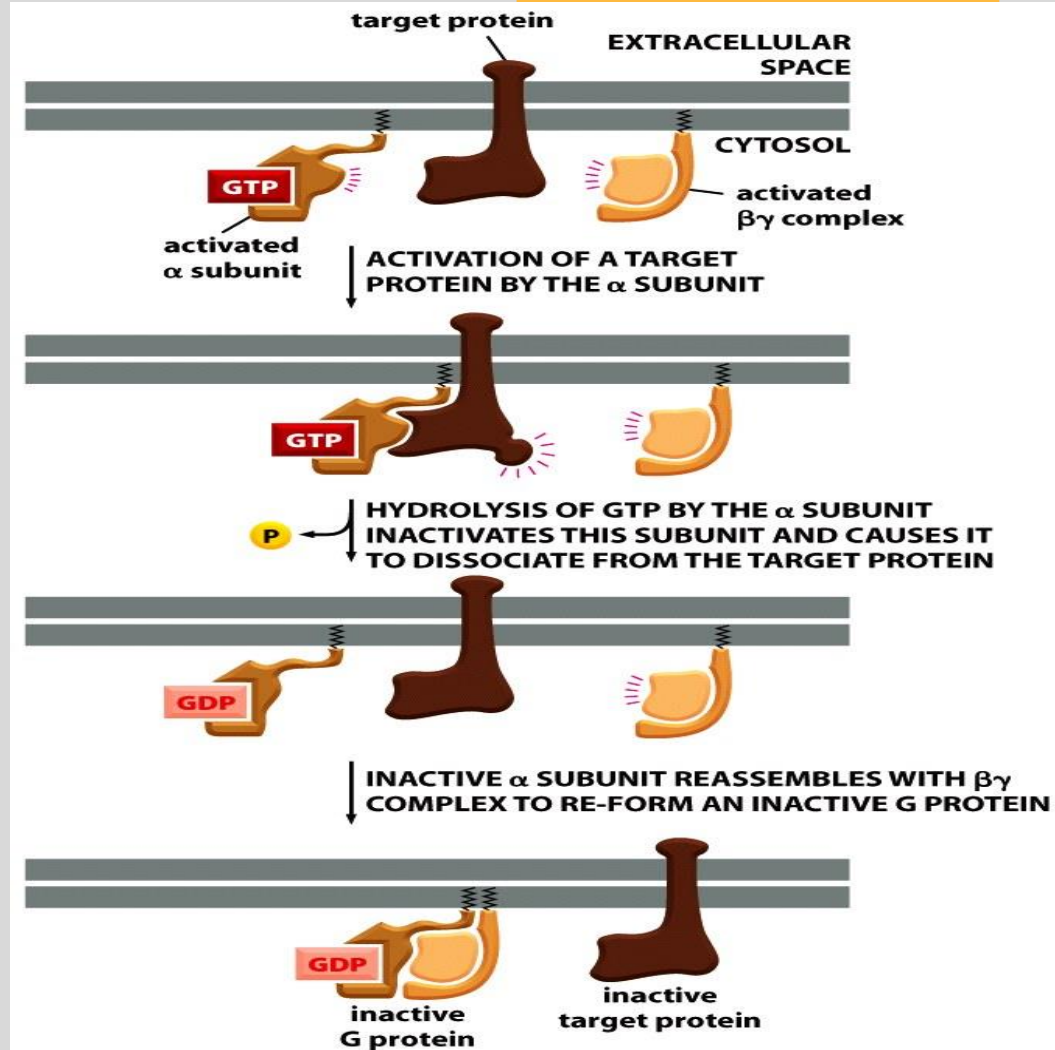
Note: α and γ subunits are still attached to the plasma membrane by lipid anchors, moving freely laterally to interact with other membrane proteins.



G Protein: Activation & Inactivation

6. $G\alpha$ and $\beta\gamma$ each activate target proteins
7. Hydrolysis of GTP by $G\alpha$'s intrinsic GTPase activity
8. Inactive $G\alpha$ reassociates with $\beta\gamma$

The activation cycle ends when α hydrolyzes the GTP and re-associates with $\beta\gamma$ to restore the inactive configuration. In this way, G proteins work like a switch — turned on or off by signal-receptor interactions on the cell's surface.



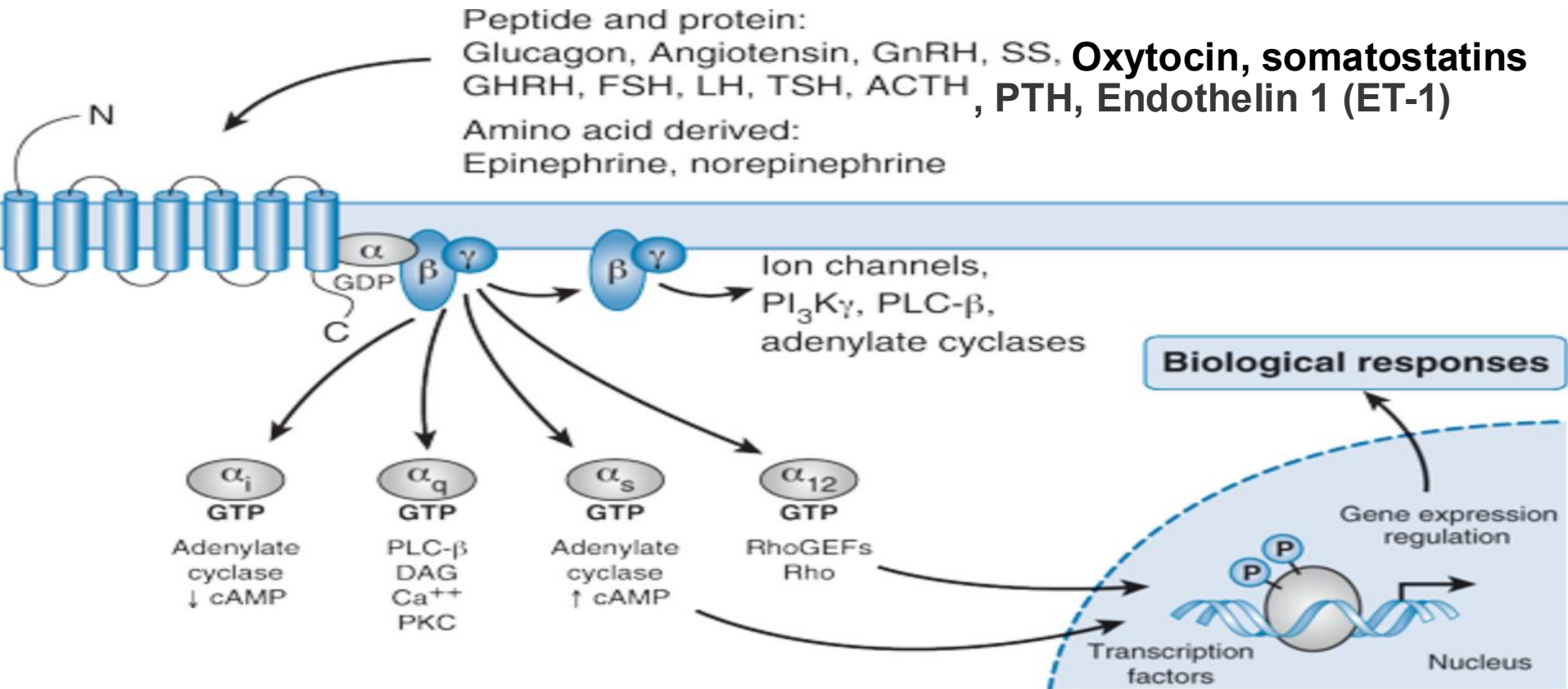
G-proteins - Subunit Isoforms

α Isoforms: at least 16 different isoforms are known in mammalian tissues, each specific to a different receptor and activating specific downstream effector(s).

$\beta\gamma$ isoforms: there are at least 5 different β and 12 different γ subunits in mammalian tissue.

Implications: The 16 α , 5 β , and 12 γ subunits can potentially form several hundred different combinations, making G proteins ideally suited to link a diversity of receptors to a diversity of effectors.

Some hormones that function through GPCR

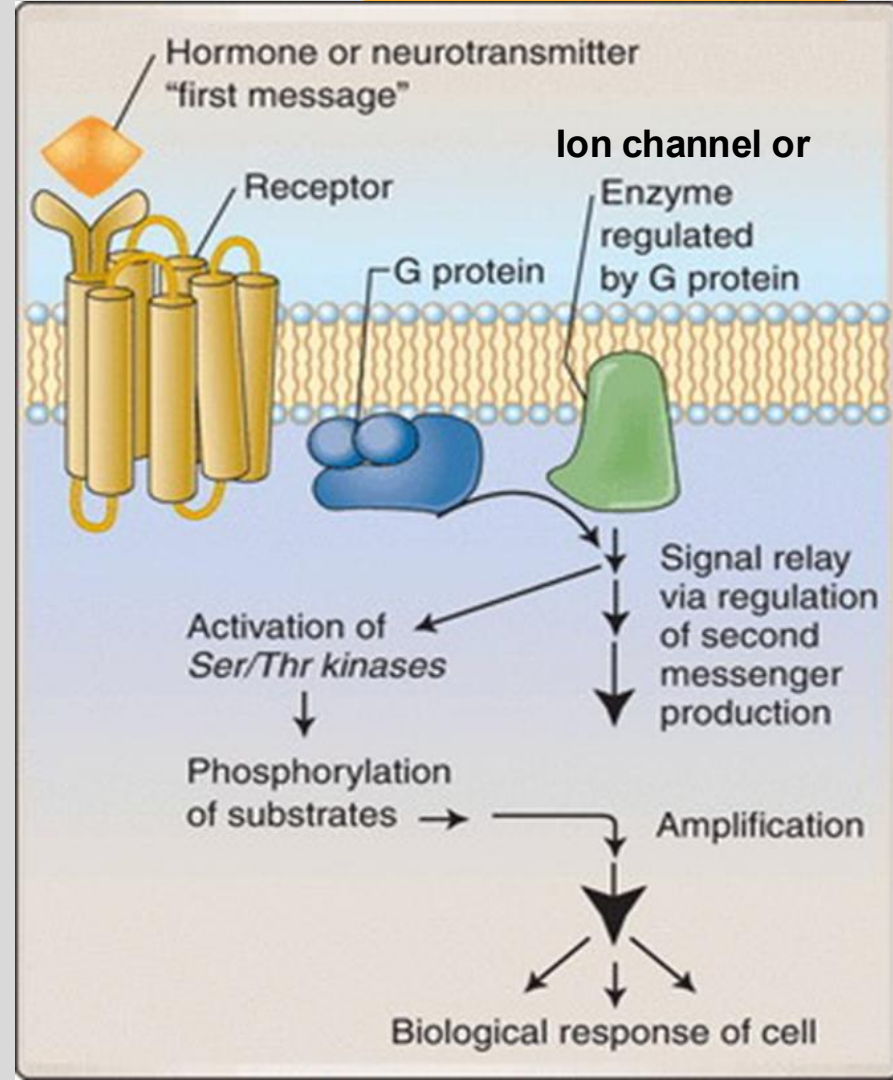


Transduction pathways

Downstream of G Protein

G proteins are **activated by a receptor**, and their free α -GTP and free $\beta\gamma$ subunits **carry “the message” to, and regulate**, a specific set of target proteins in the plasma membrane including:

- Ion channels
- Membrane-bound enzymes



G proteins that regulate ion channels

G α i and G $\beta\gamma$ control an ion channel in the heart

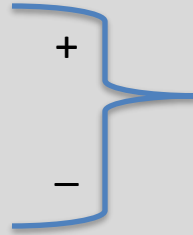
- Acetylcholine (ACh) from the vagus nerve acts on pacemaker cells, causing decreased heart rate.
- This is mediated by *muscarinic* ACh receptors, which are 7-pass receptors that work through G proteins.
- This GPCR activates G α i and liberates G $\beta\gamma$ subunits, which bind to a plasma membrane K⁺ channel, open it, increase K⁺ efflux from myocytes, hyperpolarize the myocytes, and make them harder to “fire”.
- The K⁺ channel recloses when G α i inactivates itself by hydrolyzing its bound GTP, returning to its inactive state in which the beta-gamma subunit is re-associated with the alpha subunit.

G proteins that regulate membrane-bound enzymes to produce second messengers

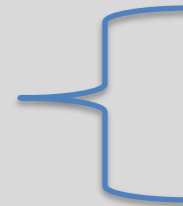
Enzymes

- $G_{\alpha s}$ or G_s : stimulates adenylyl cyclase to produce cAMP
- $G_{\alpha i}$ or G_i : inhibits adenylyl cyclase activity to reduce cAMP
- $G_{\alpha q}$ or G_q : stimulates phospholipase C to produce the IP_3 and DAG.
- $G_{\alpha o}$ or G_o : activates phospholipase A2 to produce arachidonic acid (AA)

2nd messengers



cyclic AMP (cAMP)



IP_3 (inositol trisphosphate)

Ca^{2+} (intracellular)

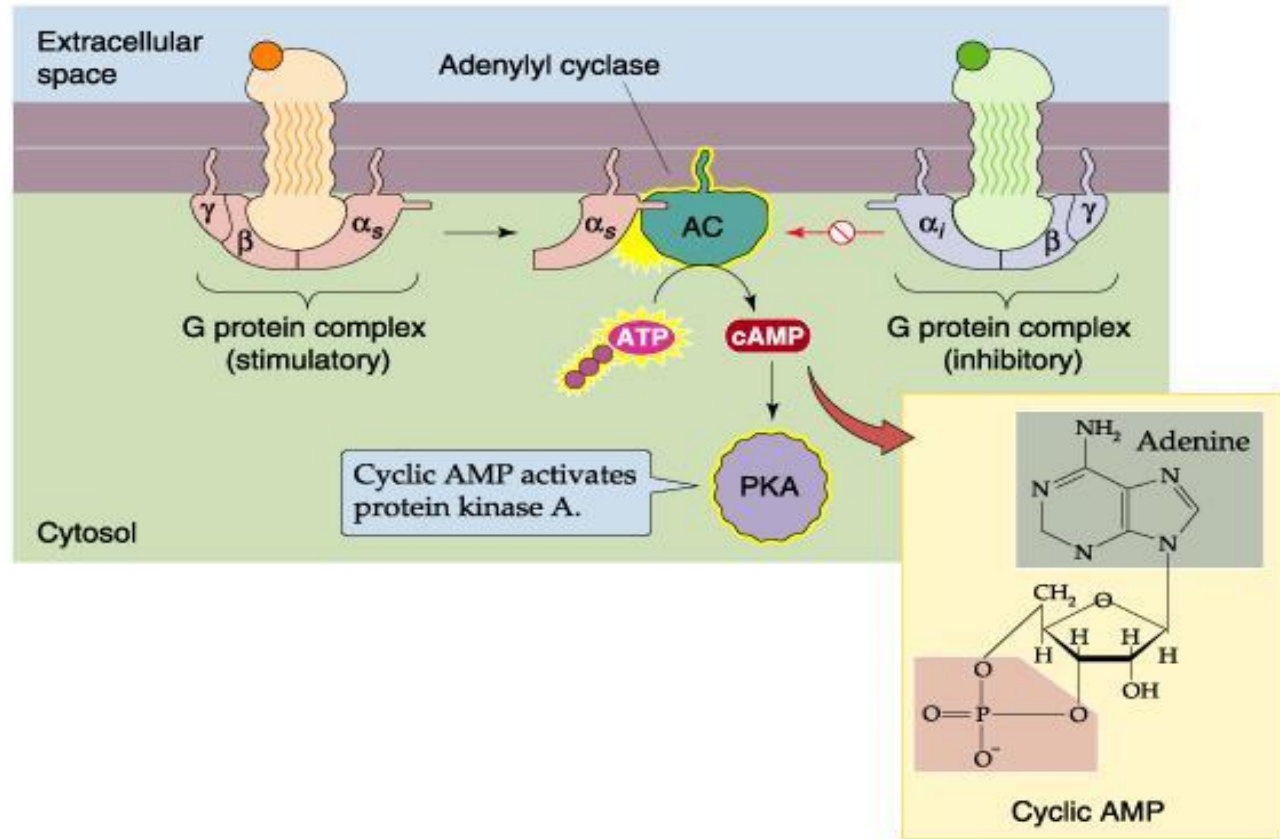
DAG (diacylglycerol)



Arachidonic acid and Eicosanoids

G_s stimulates
while G_i
inhibits
adenylyl
cyclase
activity

A G PROTEINS ACTING VIA ADENYLYL CYCLASE

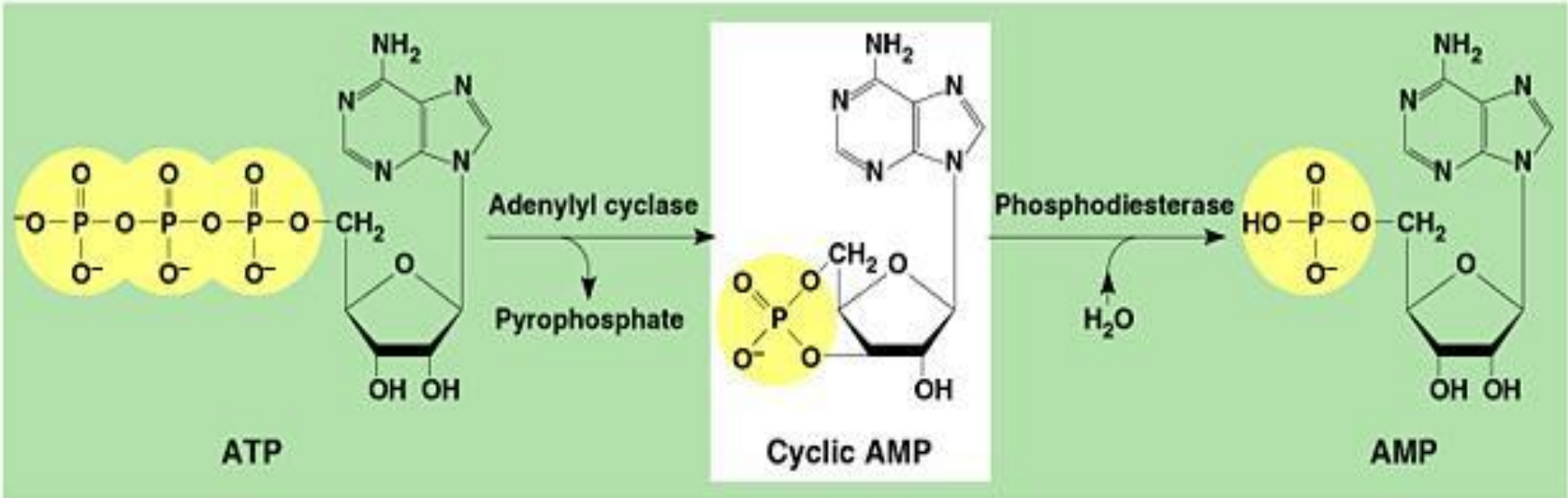


Bacterial toxins that target Gs and Gi proteins

- Cholera toxin causes “ADP ribosylation” of the $G_{\alpha s}$, which abolishes the GTPase activity of $G_{\alpha s}$, locking it in the active state in which it continuously stimulates adenylyl cyclase, causing excessive outflow of Cl^- and water into the gut.
- Pertussis toxin ADP ribosylates $G_{\alpha i}$, which inhibits the function of αi in lung airway cells, leaving it unable to inhibit adenylyl cyclase, leading to overproduction of cAMP which stimulates coughing.

Second Messenger: **Cyclic AMP (cAMP)**

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**cAMP cycle: GPCR→Gs→
adenylyl cyclase→cAMP**

Phosphodiesterase (PDE): cAMP to 5'-AMP

1. Terminate cAMP signal
2. Lead to cAMP compartmentalization

Examples of cAMP-mediated Responses

TABLE 16-3 SOME CELL RESPONSES MEDIATED BY CYCLIC AMP

EXTRACELLULAR SIGNAL MOLECULE*	TARGET TISSUE	MAJOR RESPONSE
Adrenaline (Epinephrine)	heart	increase in heart rate and force of contraction
Adrenaline	skeletal muscle	glycogen breakdown
Adrenaline, ACTH, glucagon	fat	fat breakdown
ACTH	adrenal gland	cortisol secretion

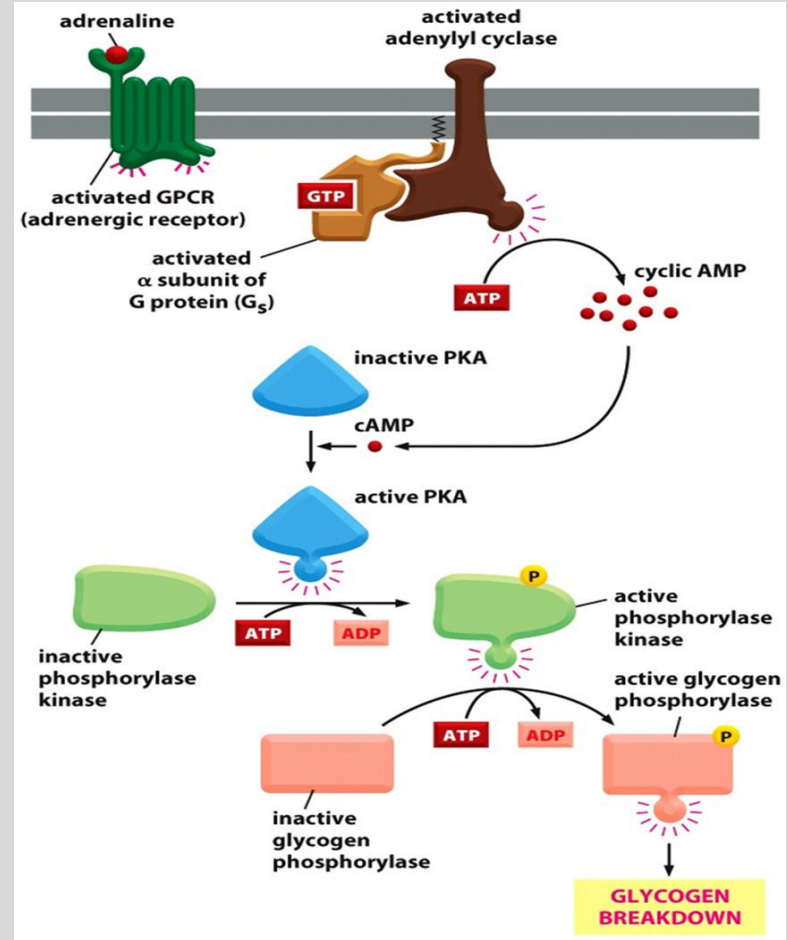
***Although all of the signal molecules listed here are hormones, some responses to local mediators and to neurotransmitters are also mediated by cyclic AMP.**

Quick response: cAMP Mediates Glycogen breakdown

Epinephrine promotes glycogen breakdown via cAMP-PKA in skeletal muscle and liver:

1. activates GPCR,
2. increases cAMP,
3. activates PKA ,
4. activates glycogen phosphorylase,
5. glycogen breakdown

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Slow response: cAMP affects gene expression

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Terminology: CRE(cyclic AMP response element);
CREB: CRE binding protein; CBP: CREB binding protein

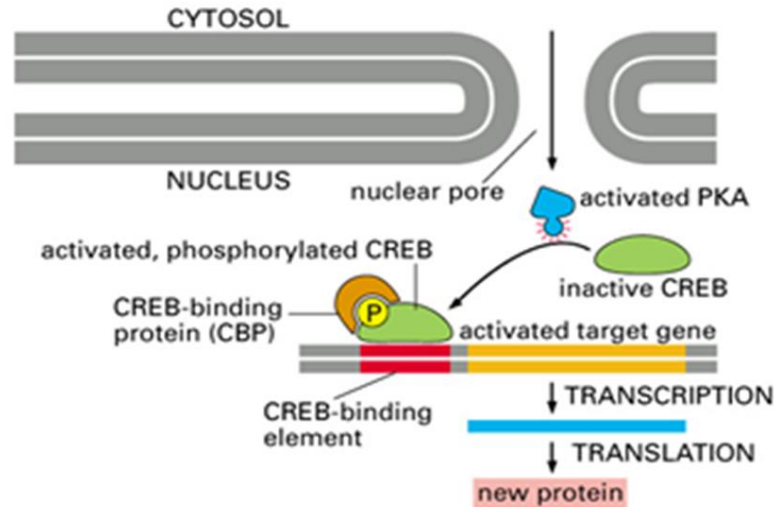
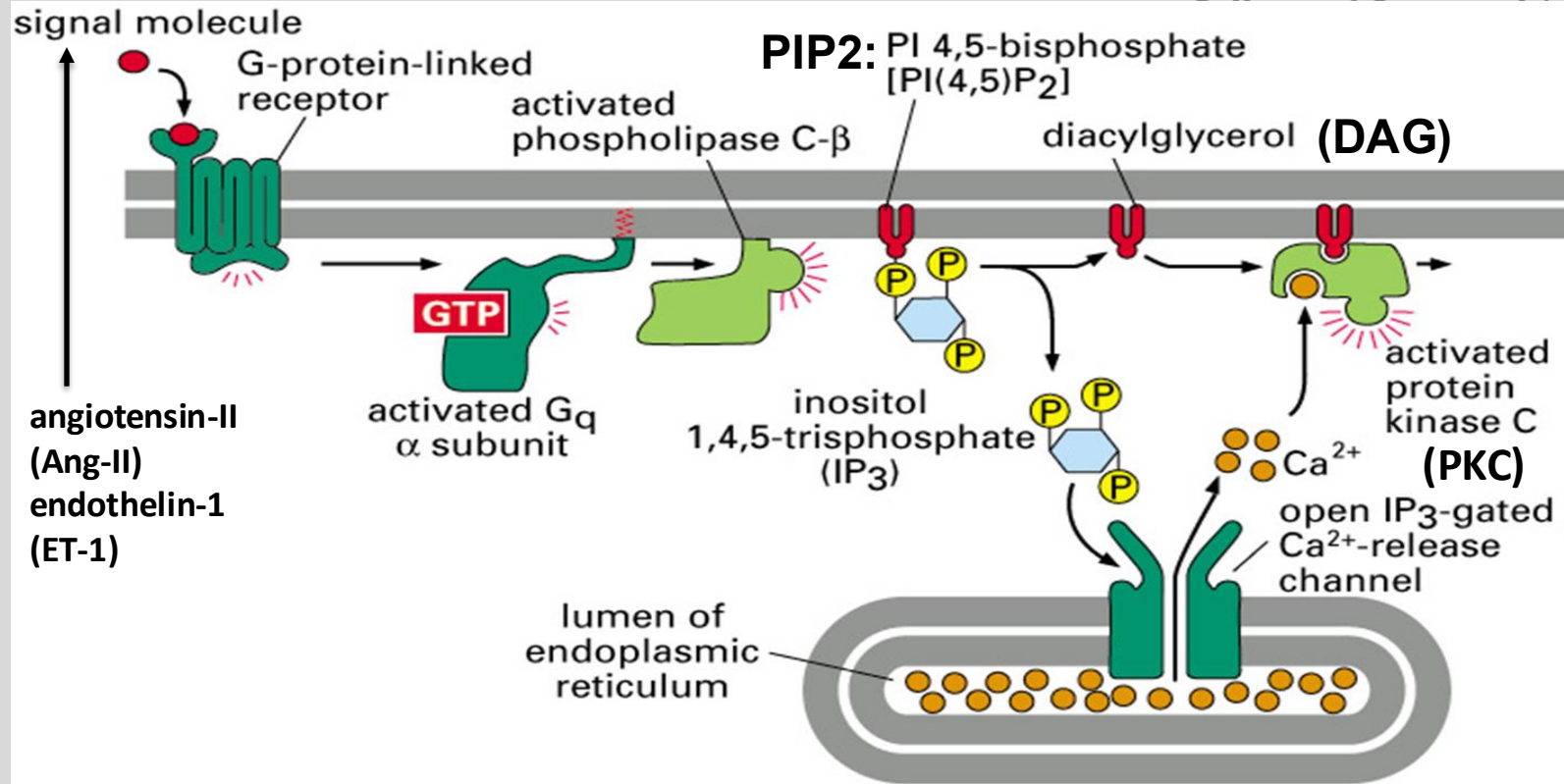


Figure 15-33 part 2 of 2. Molecular Biology of the Cell, 4th Edition.

Gq activates phospholipase C to produce IP3, DAG and Ca²⁺

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Action of IP_3

- IP_3 receptor on the endoplasmic reticulum (ER) is a calcium (Ca^{++})-specific ion channel.
- IP_3 binds to IP_3 receptors which trigger the opening of the Ca^{++} channel.
- Ca^{++} stored inside the ER rushes out into the cytosol through the open channels.
- The increased Ca^{++} signals to other proteins, participating in many cellular processes including the activation of PKC.

Action of DAG

- DAG is a lipid that remains membrane associated after it's produced by PLC- β .
- DAG helps recruit and activate a particular protein kinase, protein kinase C (PKC), which translocate from the cytosol to the plasma membrane.
- Activated PKC phosphorylates other proteins and alters their functional state.

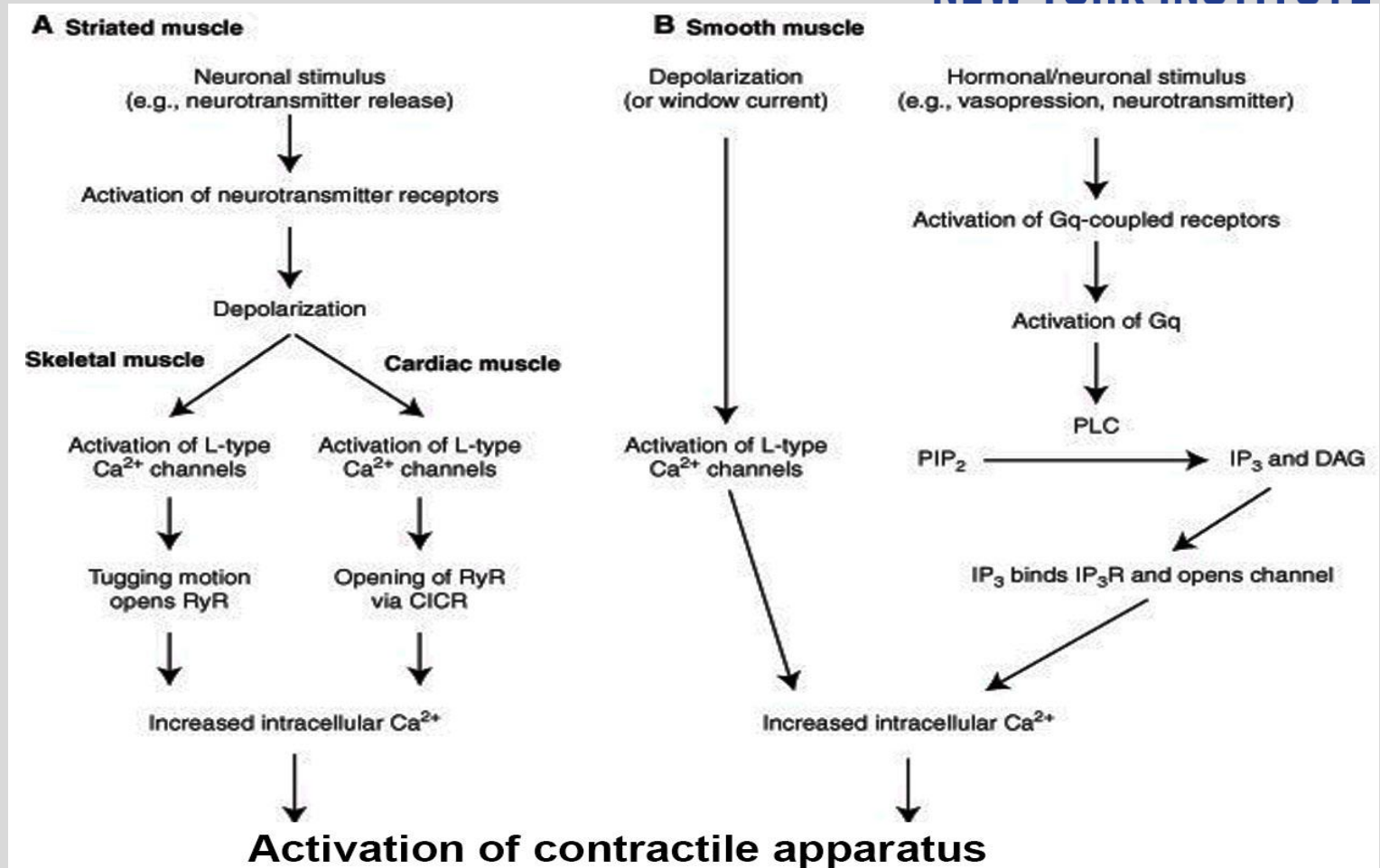
Second Messenger: Calcium

- The Ca^{++} can be easily moved around, making it a versatile second messenger in many pathways including muscle contraction, blood coagulation, and nerve function.
- Ca^{++} is low (nM) in the cytoplasm but high outside the cell (mM) and in the ER and mitochondria within the cell.
- To raise cytoplasmic Ca^{++} , the cell opens a channel which can be ligand-gated, or G protein-linked, or voltage-dependent.
- Ca^{++} interacts with various Ca^{++} responsive proteins to affect cellular functions.
- To remove Ca^{++} and terminate Ca^{++} signal in the cytoplasm, the cell uses one or more active transport mechanisms, including Ca-ATPases and Na/Ca exchangers.

Ca⁺⁺ mediates the contraction of all types of muscles

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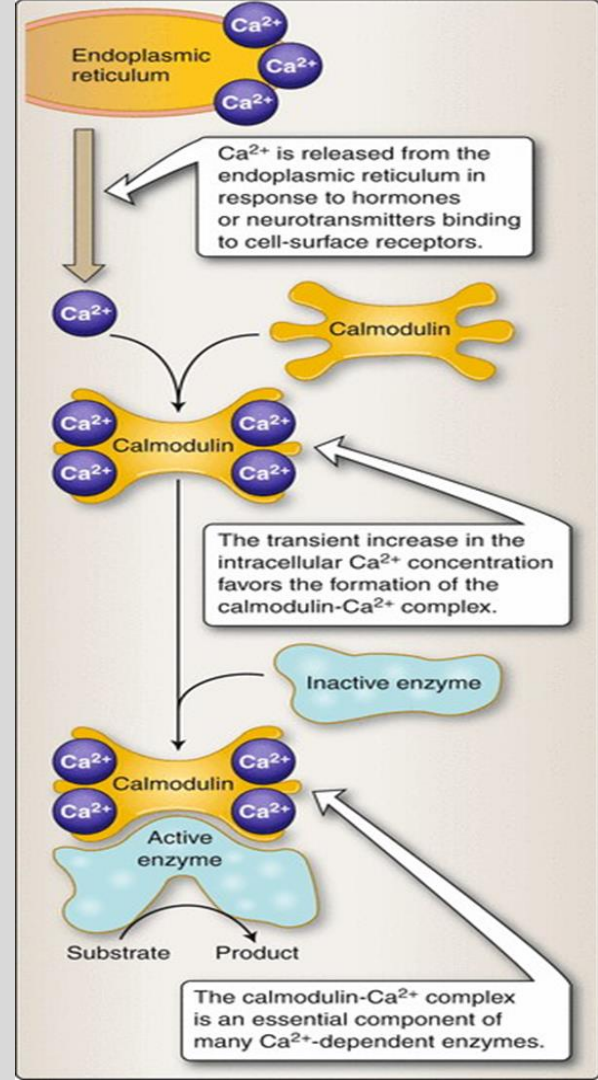
Ca⁺⁺ responsive protein: Calmodulin

In some cases, Ca exerts its effects directly (as in muscle). More often, it works in conjunction with **calmodulin**, a ubiquitous calcium-binding protein.

Calmodulin is the most widespread and common member of a family of Ca-binding proteins that includes troponin C in the muscle.

Ca-calmodulin can bind to and activate a multitude of other proteins including:

- 5 different Ca⁺⁺/calmodulin-dependent protein kinases (**CaM-kinases**).
- Myosin Light Chain Kinase (MLCK).

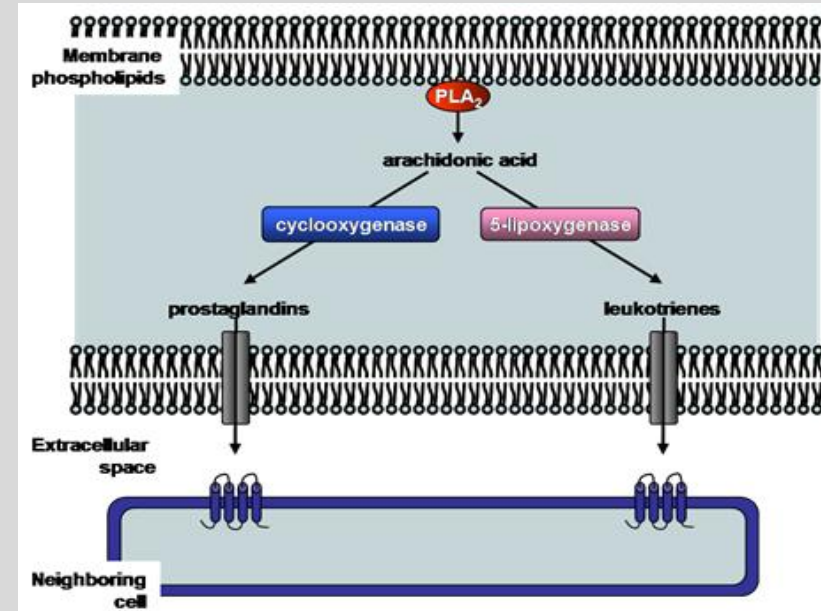


G_o activates membrane-bound phospholipase A2 to produce second messengers: Arachidonic Acid & Eicosanoids

- Agonist binds a specific GPCR;
- Receptor activates a G-protein of the G_o or G_i/G_o family;
- Activated G-protein then activates membrane-bound phospholipase A2 (PLA₂);
- PLA₂ hydrolyzes specific phospholipids to release **Arachidonic acid (AA)**.
- Enzymes in three different pathways catalyze the stereospecific insertion of oxygen into AA to produce several different **eicosanoids**.
- AA and the eicosanoids can either act as intracellular second messengers or diffuse out of the cell to exert powerful effects on processes such as inflammatory and allergic responses, platelet aggregation, vascular smooth muscle and the vascular endothelium, and gastric acid secretion.

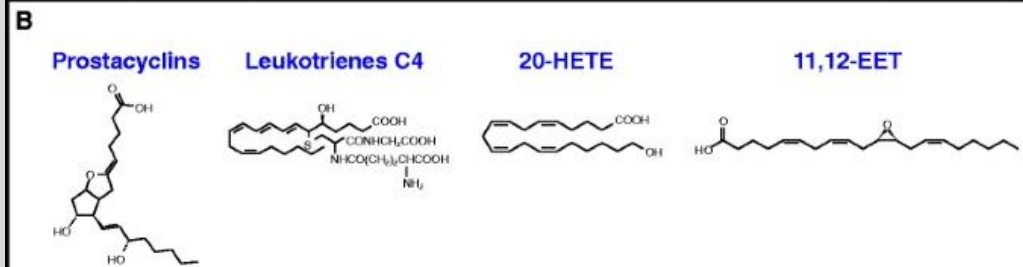
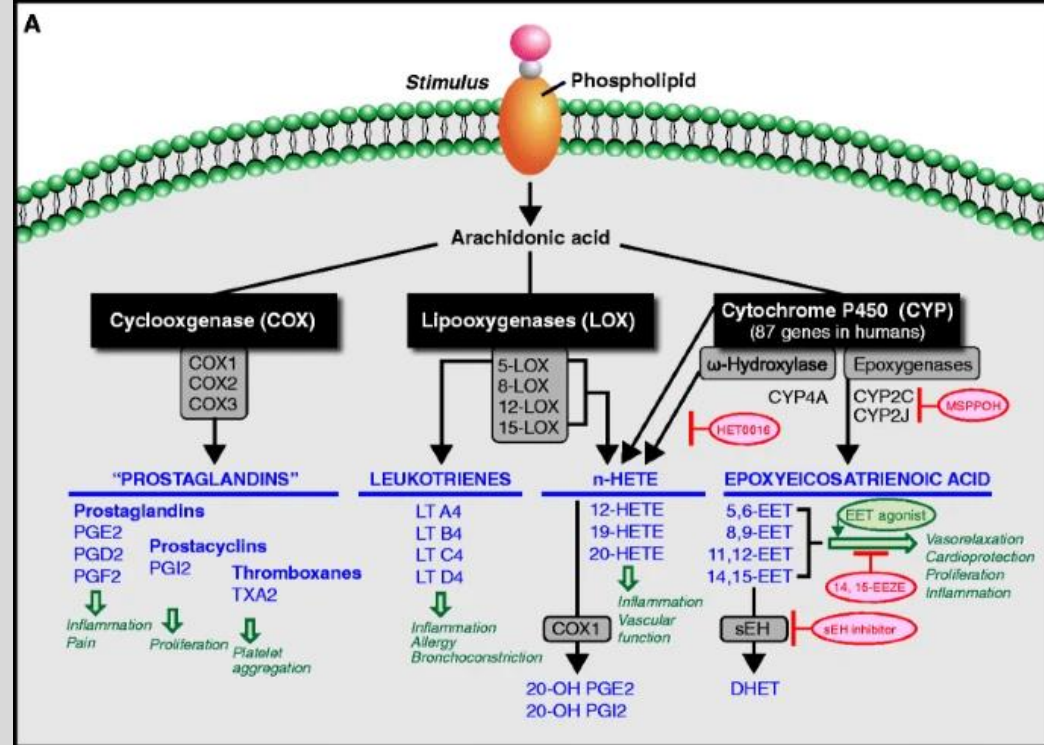
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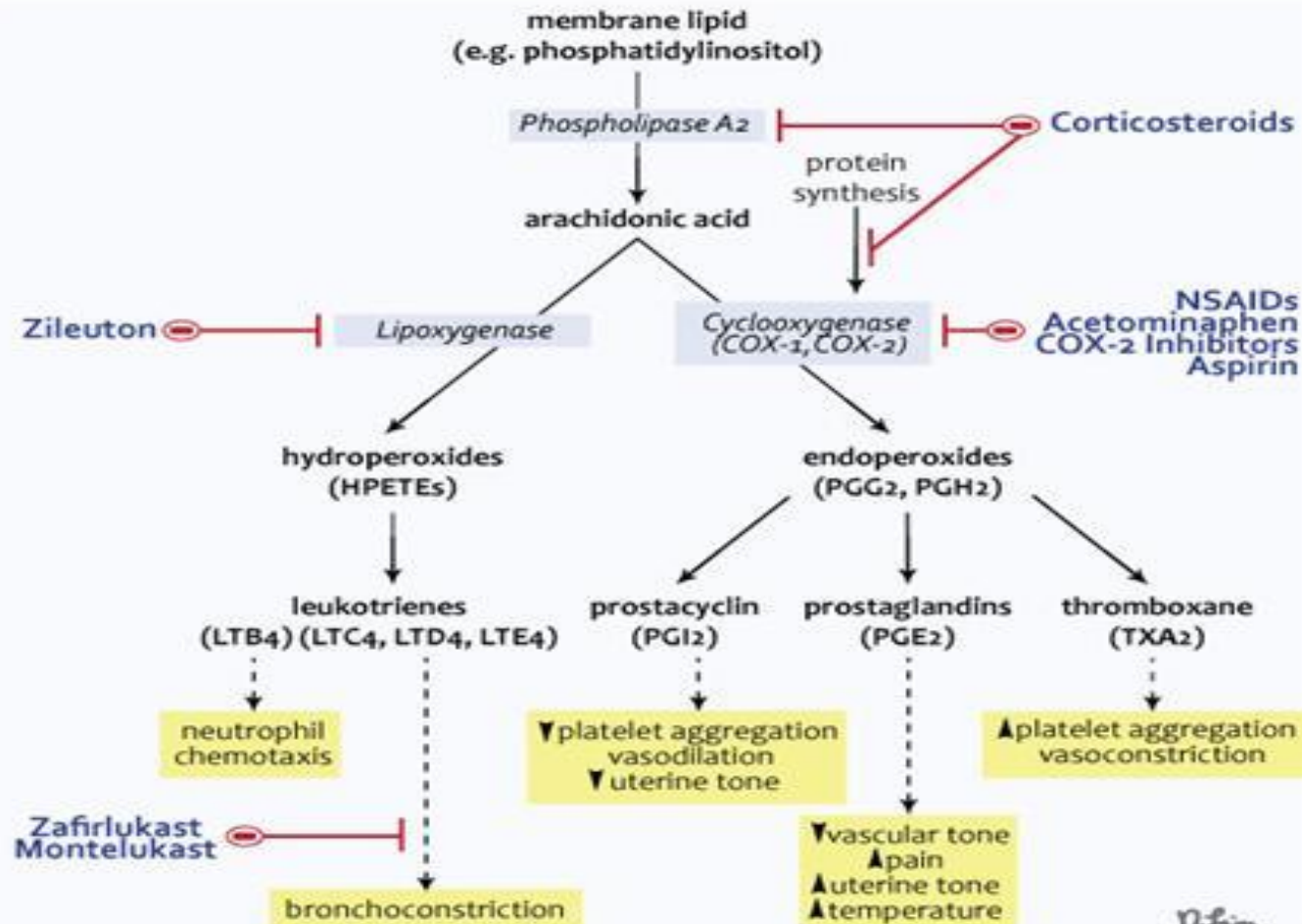


Enzymes which synthesize eicosanoids from AA

- **Cyclooxygenase enzymes (COX1 & 2)** produce prostaglandins, prostacyclins, and thromboxanes. Aspirin inhibits COX1 & 2.
- **5-lipoxygenase** enzymes produce leukotrienes and some “HETE” compounds (hydroxyeicosatetraenoic acid).
- **Epoxygenase** enzymes (Cytochrome P450) produce other HETE molecules and “EET” compounds (cis-epoxy-eicosatrienoic acid, anti-inflammation).



Arachidonic Acid Pathway



Summary of the Session

1. An ion channel-linked receptor acts as a gate that is opened by its agonist: nicotinic Ach, IP3 receptor, Ryanodine receptor, and others.
2. A GPCR activates a specific G protein by promoting GTP binding, which in turn activates a specific pathway to produce one or more second messengers.
3. Common second messengers of GPCR:
 - Gs and Gi activates or inhibits adenylyl cyclase to affect cAMP production
 - Gq activates Phospholipase C to produce IP3, DAG and Ca⁺⁺
4. G_o activates PLA2 to produce Arachidonic Acid and Eicosanoids
 - COX1 & 2: prostaglandins, prostacyclins, and thromboxanes.
 - 5-lipoxygenase: leukotrienes
 - Epoxygenase: HETE and EET compounds

Question 1

A researcher is studying signal transduction mechanisms in myocytes. She focuses on a class of heterotrimeric GTP-binding proteins that are located on the inner surface of the plasma membrane. These proteins are activated when a ligand binds to a G protein–coupled receptor (GPCR), leading to GDP-GTP exchange on the α -subunit. Once activated, these proteins dissociate into α and $\beta\gamma$ subunits, which interact with downstream enzymes or ion channels. Which of the following best describes the role of these large G proteins in cell signaling?

- a) Act as receptors for water-soluble agonists.
- b) Are activated by protein kinase A.
- c) Are activated by protein kinase C.
- d) Carry the message from a receptor to downstream effectors.
- e) Cause activation of a receptor and increase affinity of agonist binding.

Question 2

A 30-year-old man with bipolar disorder is treated with lithium. His psychiatrist explains that lithium's therapeutic effect involves inhibition of a signaling pathway that reduces the overproduction of intracellular second messengers, including inositol trisphosphate (IP_3) and diacylglycerol (DAG), in certain central nervous system cells. Lithium most likely exerts its mood-stabilizing effect by inhibiting which of the following?

- A. Adenylyl cyclase activity.
- B. $G\alpha_s$ protein function.
- C. Phospholipase C activity.
- D. PKA activity.
- E. Tyrosine kinase activity.

Other suggested resources:

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- Scholar Rx Bricks
 - Cell Signaling: <https://exchange.scholarrx.com/brick/cell-signaling>
 - Homeostatic Mechanisms: <https://exchange.scholarrx.com/brick/homeostatic-mechanisms>
- Alberts et al. *Essential Cell Biology*, 6th edition (2023), Chapter 16, “Cell Signaling” pp. 553-593 (hard copy in the library)
- Pollard and Earnshaw *Cell Biology*, Elsevier; 4th edition (2023), Section VII: Chapters 25-28, “Signaling Mechanisms” pp.429-506 .
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20180054031>
- Lippincott's Illustrated Reviews: Cell and Molecular Biology, Wolters Kluwer, 3e (2023), Unit IV: Chapters 17-19, “Cell Signaling” pp.179-206.
<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3244>

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